

第三节多元复合函数的求导法则

一、多元复合函数求导的链式法则

二、多元复合函数的全微分形式不变性

一元复合函数 $y = f(u), u = \varphi(x)$

求导法则 $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$

微分法则 $dy = f'(u)du = f'(u)\varphi'(x)dx$

一、链式法则

定理 如果函数 $u = \phi(t)$ 及 $v = \psi(t)$ 都在点 t 可导, 函数 $z = f(u, v)$ 在对应点 (u, v) 具有连续偏导数, 则复合函数 $z = f[\phi(t), \psi(t)]$ 在对应点 t 可导, 且其导数可用下列公式计算:

$$\frac{dz}{dt} = \frac{\partial z}{\partial u} \frac{du}{dt} + \frac{\partial z}{\partial v} \frac{dv}{dt}.$$

证 设 t 获得增量 Δt ,

则 $\Delta u = \phi(t + \Delta t) - \phi(t)$, $\Delta v = \psi(t + \Delta t) - \psi(t)$;

由于函数 $z = f(u, v)$ 在点 (u, v) 有连续偏导数

$$\Delta z = \frac{\partial z}{\partial u} \Delta u + \frac{\partial z}{\partial v} \Delta v + \varepsilon_1 \Delta u + \varepsilon_2 \Delta v,$$

当 $\Delta u \rightarrow 0$, $\Delta v \rightarrow 0$ 时, $\varepsilon_1 \rightarrow 0$, $\varepsilon_2 \rightarrow 0$

$$\frac{\Delta z}{\Delta t} = \frac{\partial z}{\partial u} \cdot \frac{\Delta u}{\Delta t} + \frac{\partial z}{\partial v} \cdot \frac{\Delta v}{\Delta t} + \varepsilon_1 \frac{\Delta u}{\Delta t} + \varepsilon_2 \frac{\Delta v}{\Delta t}$$

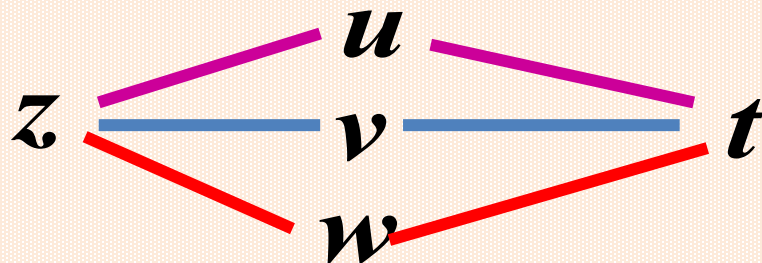
当 $\Delta t \rightarrow 0$ 时, $\Delta u \rightarrow 0$, $\Delta v \rightarrow 0$

$$\frac{\Delta u}{\Delta t} \rightarrow \frac{du}{dt}, \quad \frac{\Delta v}{\Delta t} \rightarrow \frac{dv}{dt},$$

$$\frac{dz}{dt} = \lim_{\Delta t \rightarrow 0} \frac{\Delta z}{\Delta t} = \frac{\partial z}{\partial u} \cdot \frac{du}{dt} + \frac{\partial z}{\partial v} \cdot \frac{dv}{dt}.$$

上定理的结论也可推广到中间变量多于两个的情况.

如
$$\frac{dz}{dt} = \frac{\partial z}{\partial u} \frac{du}{dt} + \frac{\partial z}{\partial v} \frac{dv}{dt} + \frac{\partial z}{\partial w} \frac{dw}{dt}$$



以上公式中的导数 $\frac{dz}{dt}$ 称为**全导数**.

说明: 若定理中 $f(u, v)$ 在点 (u, v) 偏导数连续减弱为偏导数存在, 则定理结论不一定成立.

例如: $z = f(u, v) = \begin{cases} \frac{u^2 v}{u^2 + v^2}, & u^2 + v^2 \neq 0 \\ 0, & u^2 + v^2 = 0 \end{cases}$

$$u = t, \quad v = t$$

易知: $\left. \frac{\partial z}{\partial u} \right|_{(0,0)} = f'_u(0,0) = 0, \quad \left. \frac{\partial z}{\partial v} \right|_{(0,0)} = f'_v(0,0) = 0$

但复合函数 $z = f(t, t) = \frac{t}{2}$

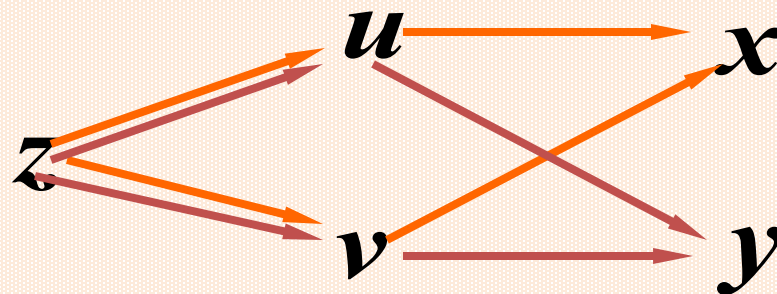
$$\frac{dz}{dt} = \frac{1}{2} \neq \frac{\partial z}{\partial u} \cdot \frac{du}{dt} + \frac{\partial z}{\partial v} \cdot \frac{dv}{dt} = 0 \cdot 1 + 0 \cdot 1 = 0$$

上定理还可推广到中间变量不是一元函数而是多元函数的情况： $z = f[\phi(x, y), \psi(x, y)]$.

如果 $u = \phi(x, y)$ 及 $v = \psi(x, y)$ 都在点 (x, y) 具有对 x 和 y 的偏导数，且函数 $z = f(u, v)$ 在对应点 (u, v) 具有连续偏导数，则复合函数 $z = f[\phi(x, y), \psi(x, y)]$ 在对应点 (x, y) 的两个偏导数存在，且可用下列公式计算

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial x}, \quad \frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial y}.$$

链式法则如图示



$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x},$$

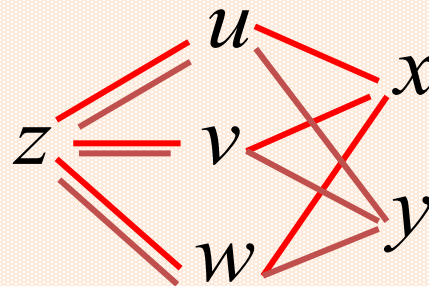
$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y}.$$

口诀: 分段用乘, 分叉用加, 单路全导, 叉路偏导.

类似地再推广，设 $u = \phi(x, y)$ 、 $v = \psi(x, y)$ 、 $w = w(x, y)$ 都在点 (x, y) 具有对 x 和 y 的偏导数，复合函数 $z = f[\phi(x, y), \psi(x, y), w(x, y)]$ 在对应点 (x, y) 两个偏导数存在，且可用下列公式计算

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial x} + \frac{\partial z}{\partial w} \frac{\partial w}{\partial x},$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial y} + \frac{\partial z}{\partial w} \frac{\partial w}{\partial y}.$$



例 1 设 $z = e^u \sin v$, 而 $u = xy$, $v = x + y$,

求 $\frac{\partial z}{\partial x}$ 和 $\frac{\partial z}{\partial y}$.

解

$$\begin{aligned}\frac{\partial z}{\partial x} &= \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x} \\ &= e^u \sin v \cdot y + e^u \cos v \cdot 1 = e^u (y \sin v + \cos v), \\ \frac{\partial z}{\partial y} &= \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y} \\ &= e^u \sin v \cdot x + e^u \cos v \cdot 1 = e^u (x \sin v + \cos v).\end{aligned}$$

特殊地 $z = f(u, x, y)$ 其中 $u = \phi(x, y)$

即 $z = f[\phi(x, y), x, y]$, 令 $v = x$, $w = y$,

$$\frac{\partial v}{\partial x} = 1, \quad \frac{\partial w}{\partial x} = 0, \quad \frac{\partial v}{\partial y} = 0, \quad \frac{\partial w}{\partial y} = 1.$$

$$\frac{\partial z}{\partial x} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial f}{\partial x},$$

$$\frac{\partial z}{\partial y} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial f}{\partial y}.$$

区别类似

两者的区别

把复合函数 $z = f[\phi(x, y), x, y]$ 中的 y 看作不变而对 x 的偏导数

把 $z = f(u, x, y)$ 中的 u 及 y 看作不变而对 x 的偏导数

例2. $u = f(x, y, z) = e^{x^2+y^2+z^2}$, $z = x^2 \sin y$, 求 $\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial y}$

解: $\frac{\partial u}{\partial x} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial z} \cdot \frac{\partial z}{\partial x}$

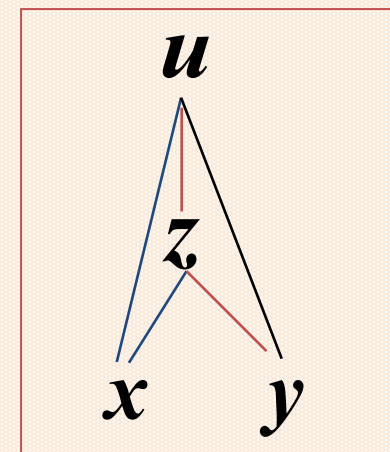
$$= 2xe^{x^2+y^2+z^2} + 2ze^{x^2+y^2+z^2} \cdot 2x \sin y$$

$$= 2x(1 + 2x^2 \sin^2 y)e^{x^2+y^2+x^4 \sin^2 y}$$

$$\frac{\partial u}{\partial y} = \frac{\partial f}{\partial y} + \frac{\partial f}{\partial z} \cdot \frac{\partial z}{\partial y}$$

$$= 2ye^{x^2+y^2+z^2} + 2ze^{x^2+y^2+z^2} \cdot x^2 \cos y$$

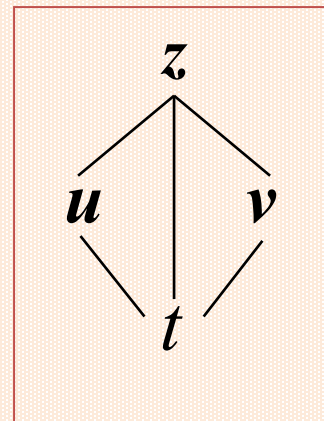
$$= 2(y + x^4 \sin y \cos y)e^{x^2+y^2+x^4 \sin^2 y}$$



例3. 设 $z = uv + \sin t$, $u = e^t$, $v = \cos t$, 求全导数 $\frac{dz}{dt}$.

解:

$$\begin{aligned}\frac{dz}{dt} &= \frac{\partial z}{\partial u} \cdot \frac{du}{dt} + \frac{\partial z}{\partial v} \cdot \frac{dv}{dt} + \frac{\partial z}{\partial t} \\ &= v e^t - u \sin t + \cos t \\ &= e^t (\cos t - \sin t) + \cos t\end{aligned}$$



注意: 多元抽象复合函数求导在偏微分方程变形与验证解的问题中经常遇到, 下列两个例题有助于掌握这方面问题的求导技巧与常用导数符号.

例4. 设 $w = f(x + y + z, xyz)$, f 具有二阶连续偏导数,

求 $\frac{\partial w}{\partial x}, \frac{\partial^2 w}{\partial x \partial z}$.

解: 令 $u = x + y + z, v = xyz$, 则

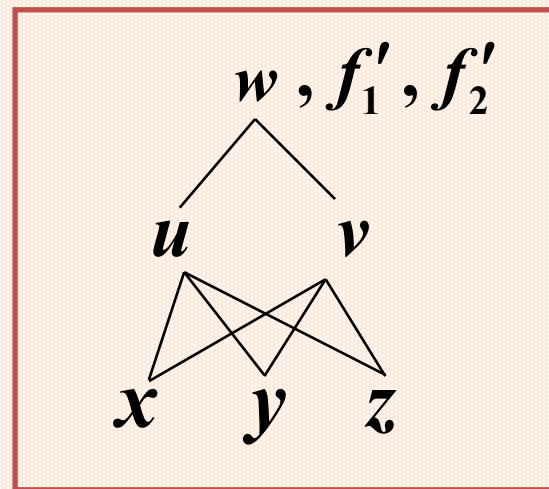
$$w = f(u, v)$$

$$\frac{\partial w}{\partial x} = f'_1 \cdot 1 + f'_2 \cdot yz$$

$$= \underline{f'_1(x + y + z, xyz)} + \underline{yz f'_2(x + y + z, xyz)}$$

$$\frac{\partial^2 w}{\partial x \partial z} = f''_{11} \cdot 1 + f''_{12} \cdot xy + y f'_2 + yz [f''_{21} \cdot 1 + f''_{22} \cdot xy]$$

$$= f''_{11} + y(x + z)f''_{12} + xy^2z f''_{22} + y f'_2$$



例5. 设 $u = f(x, y)$ 二阶偏导数连续, 求下列表达式在极坐标系下的形式 $(\frac{\partial u}{\partial x})^2 + (\frac{\partial u}{\partial y})^2$

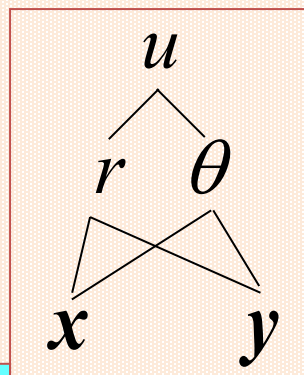
解: 已知 $x = r \cos \theta$, $y = r \sin \theta$, 则

$$r = \sqrt{x^2 + y^2}, \quad \theta = \arctan \frac{y}{x}$$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial u}{\partial \theta} \frac{\partial \theta}{\partial x}$$

$$\frac{\partial r}{\partial x} = \frac{x}{r}, \quad \frac{\partial \theta}{\partial x} = \frac{\frac{-y}{x^2}}{1 + (\frac{y}{x})^2} = \frac{-y}{x^2 + y^2}$$

$$= \frac{\partial u}{\partial r} \frac{x}{r} - \frac{\partial u}{\partial \theta} \frac{y}{r^2} = \frac{\partial u}{\partial r} \cos \theta - \frac{\partial u}{\partial \theta} \frac{\sin \theta}{r}$$

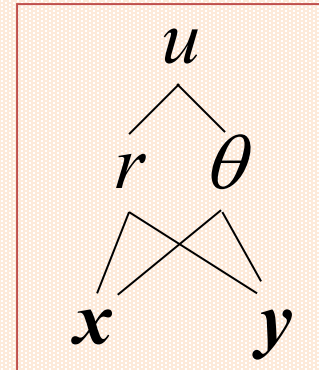


$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial u}{\partial \theta} \frac{\partial \theta}{\partial y}$$

$$\frac{\partial r}{\partial y} = \frac{y}{r}, \quad \frac{\partial \theta}{\partial y} = \frac{\frac{1}{x}}{1 + (\frac{y}{x})^2} = \frac{x}{x^2 + y^2}$$

$$\begin{aligned} &= \frac{\partial u}{\partial r} \frac{y}{r} + \frac{\partial u}{\partial \theta} \frac{x}{r^2} \\ &= \frac{\partial u}{\partial r} \sin \theta + \frac{\partial u}{\partial \theta} \frac{\cos \theta}{r} \end{aligned}$$

$$\therefore \left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 = \left(\frac{\partial u}{\partial r}\right)^2 + \frac{1}{r^2} \left(\frac{\partial u}{\partial \theta}\right)^2$$



例6 设 $u = f(x - y, y - z, z - x)$, 假设 f 对其中变量有直到二阶的连续偏导数, 求 $\frac{\partial^2 u}{\partial x^2}$ 及 $\frac{\partial^2 u}{\partial y \partial z}$ (上海交大)

解 令 $t = x - y$, $v = y - z$, $w = z - x$, 则,

$$\frac{\partial u}{\partial x} = \frac{\partial f}{\partial t} \frac{\partial t}{\partial x} + \frac{\partial f}{\partial v} \frac{\partial v}{\partial x} + \frac{\partial f}{\partial w} \frac{\partial w}{\partial x} = \frac{\partial f}{\partial t} - \frac{\partial f}{\partial w},$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial t} \left(\frac{\partial f}{\partial t} - \frac{\partial f}{\partial w} \right) \frac{\partial t}{\partial x} + \frac{\partial}{\partial w} \left(\frac{\partial f}{\partial t} - \frac{\partial f}{\partial w} \right) \frac{\partial w}{\partial x} = \frac{\partial^2 f}{\partial t^2} - 2 \frac{\partial^2 f}{\partial t \partial w} + \frac{\partial^2 f}{\partial w^2},$$

$$\frac{\partial u}{\partial y} = \frac{\partial f}{\partial t} \frac{\partial t}{\partial y} + \frac{\partial f}{\partial v} \frac{\partial v}{\partial y} + \frac{\partial f}{\partial w} \frac{\partial w}{\partial y} = \frac{\partial f}{\partial v} - \frac{\partial f}{\partial t},$$

$$\frac{\partial^2 u}{\partial y \partial z} = \frac{\partial}{\partial t} \left(\frac{\partial f}{\partial v} - \frac{\partial f}{\partial t} \right) \frac{\partial t}{\partial z} + \frac{\partial}{\partial v} \left(\frac{\partial f}{\partial v} - \frac{\partial f}{\partial t} \right) \frac{\partial v}{\partial z} + \frac{\partial}{\partial w} \left(\frac{\partial f}{\partial v} - \frac{\partial f}{\partial t} \right) \frac{\partial w}{\partial z}$$

$$= \frac{\partial^2 f}{\partial t \partial v} - \frac{\partial^2 f}{\partial v^2} + \frac{\partial^2 f}{\partial v \partial w} - \frac{\partial^2 f}{\partial t \partial w}.$$

例7 设函数 $\varphi(z)$ 和 $\psi(z)$ 具有二阶连续导数,
 并设 $u = x\varphi(x+y) + y\psi(x+y)$, 试证 $\frac{\partial^2 u}{\partial x^2} - 2\frac{\partial^2 u}{\partial x\partial y} + \frac{\partial^2 u}{\partial y^2} = 0$. (中科院)

证 $\frac{\partial u}{\partial x} = \varphi(x+y) + x\varphi'(x+y) + y\psi'(x+y)$

$$\begin{array}{c} \varphi(x+y) \\ \downarrow \\ u \end{array}$$

$$\frac{\partial^2 u}{\partial x^2} = \varphi'(x+y) + \varphi'(x+y) + x\varphi''(x+y) + y\psi''(x+y)$$

$$= 2\varphi'(x+y) + x\varphi''(x+y) + y\psi''(x+y)$$

$$\frac{\partial^2 u}{\partial x\partial y} = \varphi'(x+y) + x\varphi''(x+y) + \psi'(x+y) + y\psi''(x+y)$$

$$= \varphi'(x+y) + \psi'(x+y) + x\varphi''(x+y) + y\psi''(x+y)$$

$$\begin{array}{c} \nearrow x \\ \varphi \rightarrow u \\ \searrow y \end{array}$$

同理, $\frac{\partial^2 u}{\partial y^2} = 2\psi'(x+y) + x\varphi''(x+y) + y\psi''(x+y)$

所以有 $\frac{\partial^2 u}{\partial x^2} - 2\frac{\partial^2 u}{\partial x\partial y} + \frac{\partial^2 u}{\partial y^2} = 0$.

综合题

例8 设函数 $f(u)$ 具有二阶连续导数, $z = f(e^x \cos y)$ 满足 $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = (4z + e^x \cos y)e^{2x}$, 若 $f(0) = 0, f'(0) = 0$, 求 $f(u)$ 的表达式. (2016年考研高数二)

证 设 $u = e^x \cos y$, 则, $z = f(u) = f(e^x \cos y)$

$$\frac{\partial z}{\partial x} = f'(u)e^x \cos y, \quad \frac{\partial^2 z}{\partial x^2} = f''(u)e^{2x} \cos^2 y + f'(u)e^x \cos y$$

$$\frac{\partial z}{\partial y} = -f'(u)e^x \sin y, \quad \frac{\partial^2 z}{\partial y^2} = f''(u)e^{2x} \sin^2 y - f'(u)e^x \cos y$$

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = f''(u)e^{2x} = f''(e^x \cos y)e^{2x}$$

由条件 $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = (4z + e^x \cos y)e^{2x}$

可知 $f''(u) = 4f(u) + u$,

这是一个二阶常系数线性非齐次方程. 对应齐次方程的通解为:

$f(u) = C_1 e^{2u} + C_2 e^{-2u}$, 其中 C_1, C_2 为任意常数.

对应非齐次方程特解可求得为 $\bar{y} = -\frac{1}{4}u$

故非齐次方程通解为 $f(u) = C_1 e^{2u} + C_2 e^{-2u} - \frac{1}{4}u$,

将初始条件 $f(0) = 0, f'(0) = 0$ 代入, 可得 $C_1 = \frac{1}{16}, C_2 = -\frac{1}{16}$

所以 $f(u)$ 的表达式为 $f(u) = \frac{1}{16}e^{2u} - \frac{1}{16}e^{-2u} - \frac{1}{4}u$.

二、多元复合函数的全微分的形式不变性

设函数 $z = f(u, v)$, $u = \varphi(x, y)$, $v = \psi(x, y)$ 都可微,
则复合函数 $z = f(\varphi(x, y), \psi(x, y))$ 的全微分为

$$\begin{aligned} \mathrm{d}z &= \frac{\partial z}{\partial x} \mathrm{d}x + \frac{\partial z}{\partial y} \mathrm{d}y \\ &= \left(\frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x} \right) \mathrm{d}x + \left(\frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y} \right) \mathrm{d}y \\ &= \frac{\partial z}{\partial u} \left(\frac{\partial u}{\partial x} \mathrm{d}x + \frac{\partial u}{\partial y} \mathrm{d}y \right) + \frac{\partial z}{\partial v} \left(\frac{\partial v}{\partial x} \mathrm{d}x + \frac{\partial v}{\partial y} \mathrm{d}y \right) \\ &= \frac{\partial z}{\partial u} \mathrm{d}u + \frac{\partial z}{\partial v} \mathrm{d}v \end{aligned}$$

可见无论 u, v 是自变量还是中间变量, 其全微分表达式都一样, 这性质叫做**全微分形式不变性**.

函数和、差、积、商的微分法则

$$d(u \pm v) = du \pm dv$$

$$d(Cu) = Cdu$$

$$d(uv) = vdu + udv$$

$$d\left(\frac{u}{v}\right) = \frac{vdu - u dv}{v^2}$$

微分形式的不变性

$$dz = \frac{\partial z}{\partial u} du + \frac{\partial z}{\partial v} dv$$

$$= \frac{\partial z}{\partial u} \left(\frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy \right) + \frac{\partial z}{\partial v} \left(\frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy \right)$$

$$= \left(\frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x} \right) dx + \left(\frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y} \right) dy$$

$$\frac{\partial z}{\partial x}$$

$$\frac{\partial z}{\partial y}$$

- (A) 熟练利用微分形式不变性及四则运算微分法则是一种很好的计算偏导的方法，特别是隐函数求偏导.
- (B) 利用微分形式不变性可以一次性求出两个偏导，在隐函数方程组里可以一次求出四个偏导.

例 8 已知 $e^{-xy} - 2z + e^z = 0$, 求 $\frac{\partial z}{\partial x}$ 和 $\frac{\partial z}{\partial y}$.

解 $\because d(e^{-xy} - 2z + e^z) = 0,$

$$\therefore e^{-xy}d(-xy) - 2dz + e^z dz = 0,$$

$$(e^z - 2)dz = e^{-xy}(xdy + ydx)$$

$$dz = \frac{ye^{-xy}}{(e^z - 2)}dx + \frac{xe^{-xy}}{(e^z - 2)}dy$$

$$\frac{\partial z}{\partial x} = \frac{ye^{-xy}}{e^z - 2}, \quad \frac{\partial z}{\partial y} = \frac{xe^{-xy}}{e^z - 2}.$$

例 9. 利用全微分形式不变性再解例1.

解: $dz = d(e^u \sin v)$

$$= e^u \sin v du + e^u \cos v dv$$

$$= e^{xy} [\sin(x+y) d(xy) + \cos(x+y) d(x+y)]$$

$$= e^{xy} [\sin(x+y)(ydx + xdy) + \cos(x+y)(dx + dy)]$$

$$= e^{xy} [y \sin(x+y) + \cos(x+y)] dx \\ + e^{xy} [x \sin(x+y) + \cos(x+y)] dy$$

所以 $\frac{\partial z}{\partial x} = e^{xy} [y \cdot \sin(x+y) + \cos(x+y)]$

$$\frac{\partial z}{\partial y} = e^{xy} [x \cdot \sin(x+y) + \cos(x+y)]$$

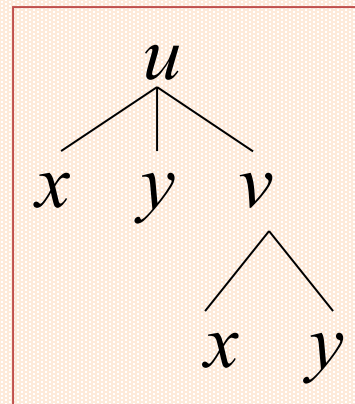
三、内容小结

1. 复合函数求导的链式法则

“分段用乘, 分叉用加, 单路全导, 叉路偏导”

例如, $u = f(x, y, v)$, $v = \varphi(x, y)$,

$$\frac{\partial u}{\partial x} = f'_1 + f'_3 \cdot \varphi'_1; \quad \frac{\partial u}{\partial y} = f'_2 + f'_3 \cdot \varphi'_2$$



2. 全微分形式不变性

对 $z = f(u, v)$, 不论 u, v 是自变量还是因变量,

$$dz = f'_u(u, v)du + f'_v(u, v)dv$$

练习

题1 $z = \arctan \frac{x}{y}, x = u + v, y = u - v$

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial v}$$

$$= \frac{1}{1 + (\frac{x}{y})^2} \cdot \frac{1}{y} \cdot 1 + \frac{1}{1 + (\frac{x}{y})^2} \cdot (-\frac{x}{y^2}) \cdot (-1)$$

$$= \frac{y + x}{x^2 + y^2} = \frac{u}{u^2 + v^2}$$

.....

题2 $u = f\left(\frac{x}{y}, \frac{y}{z}\right)$

$$\frac{\partial u}{\partial x} = f'_1 \cdot \frac{1}{y} = \frac{1}{y} f'_1$$

$$\frac{\partial u}{\partial y} = f'_1 \cdot \left(-\frac{x}{y^2}\right) + f'_2 \cdot \frac{1}{z} = -\frac{x}{y^2} f'_1 + \frac{1}{z} f'_2$$

$$\frac{\partial u}{\partial z} = f'_2 \cdot \left(-\frac{y}{z^2}\right) = -\frac{y}{z^2} f'_2$$

题 3 $z = f(u, x, y), \quad u = xe^y$

$$\frac{\partial z}{\partial x} = f'_1 \cdot e^y + f'_2$$

$$\begin{aligned} \frac{\partial^2 z}{\partial x \partial y} &= e^y \cdot f'_1 + xe^{2y} \cdot f''_{11} + e^y \cdot f''_{13} \\ &\quad + xe^y \cdot f''_{21} + f''_{23} \end{aligned}$$

提高题

1. 已知 $f(x, y)|_{y=x^2} = 1$, $f'_1(x, y)|_{y=x^2} = 2x$, 求 $f'_2(x, y)|_{y=x^2}$.

解: 由 $f(x, x^2) = 1$ 两边对 x 求导, 得

$$f'_1(x, x^2) + f'_2(x, x^2) \cdot 2x = 0$$

$$\begin{array}{c} \downarrow \\ f'_1(x, x^2) = 2x \\ f'_2(x, x^2) = -1 \end{array}$$

2、已知函数 $f(x, y)$ 具有连续二阶偏导数, 且

$$f''_{xx}(x, y) = f''_{yy}(x, y), \quad f(x, 2x) = x^2, \quad f'_x(x, 2x) = x,$$

试求 $f''_{xx}(x, 2x)$ 与 $f''_{xy}(x, 2x)$.

解 在等式 $f(x, 2x) = x^2$ 两边对 x 求导数得

$$f'_x(x, 2x) + 2f'_y(x, 2x) = 2x,$$

两边再对 x 求导数得,

$$f''_{xx}(x, 2x) + 2f''_{xy}(x, 2x) + 2f''_{yx}(x, 2x) + 4f''_{yy}(x, 2x) = 2,$$

由条件得, $5f''_{xx}(x, 2x) + 4f''_{xy}(x, 2x) = 2,$

在 $f'_x(x, 2x) = x^2$ 两边对 x 求导得

$$f''_{xx}(x, 2x) + 2f''_{xy}(x, 2x) = 1,$$

将上两式联立解得

$$f''_{xx}(x, 2x) = 0, \quad f''_{xy}(x, 2x) = \frac{1}{2}.$$

3. 设函数 $z = f(x, y)$ 在点 $(1, 1)$ 处可微, 且

$$f(1, 1) = 1, \quad \left. \frac{\partial f}{\partial x} \right|_{(1, 1)} = 2, \quad \left. \frac{\partial f}{\partial y} \right|_{(1, 1)} = 3,$$

$\varphi(x) = \underline{f(x, f(x, x))}$, 求 $\left. \frac{d}{dx} \varphi^3(x) \right|_{x=1}$ · (考研题)

解: 由题设 $\varphi(1) = f(1, f(1, 1)) = f(1, 1) = 1$

$$\begin{aligned} \left. \frac{d}{dx} \varphi^3(x) \right|_{x=1} &= 3\varphi^2(x) \left. \frac{d\varphi}{dx} \right|_{x=1} \\ &= 3 \left[\underline{f'_1(x, f(x, x))} \right. \\ &\quad \left. + \underline{f'_2(x, f(x, x))} (\underline{f'_1(x, x)} + \underline{f'_2(x, x)}) \right] \Big|_{x=1} \\ &= 3 \cdot [2 + 3 \cdot (2 + 3)] = 51 \end{aligned}$$

4、设 f 连续可导, $z(x, y) = \int_0^y e^y f(x-t)dt$, 求 $\frac{\partial^2 z}{\partial x \partial y}$.

解 对变上限积分做变换 $x-t=u$, 有

$$z(x, y) = e^y \int_{x-y}^x f(u)du,$$

于是,

$$\frac{\partial z}{\partial x} = e^y [f(x) - f(x-y)],$$

$$\frac{\partial^2 z}{\partial x \partial y} = e^y [f(x) - f(x-y)] + e^y f'(x-y).$$

$$F(u, v) = \int_u^v f(t)dt,$$

$$\frac{\partial F}{\partial u} = -f(u), \quad \frac{\partial F}{\partial v} = f(v)$$

$$z = F(u(x, y), v(x, y)) = \int_{u(x, y)}^{v(x, y)} f(t)dt,$$

$$\frac{\partial z}{\partial x} = \frac{\partial F}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial F}{\partial v} \frac{\partial v}{\partial x} = -f(u) \frac{\partial u}{\partial x} + f(v) \frac{\partial v}{\partial x}$$

$$\frac{\partial z}{\partial y} = \frac{\partial F}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial F}{\partial v} \frac{\partial v}{\partial y} = -f(u) \frac{\partial u}{\partial y} + f(v) \frac{\partial v}{\partial y}$$